



College of Basic and Applied Sciences
School of Engineering Sciences · Department of Computer Engineering

CPEN 208 — Project 2

API / Web Service implementing the Project 1 functionalities

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Course: CPEN 208 — Introduction to Software Engineering

Semester: Second Semester, 2025/2026

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Spring Boot 3.5

Java 21

Spring JDBC

JWT

BCrypt

GitHub: <https://github.com/doxaodoi/project-1-2>

1. Overview

Project 2 is an **API / web service** that implements the Project 1 functionalities and exposes them over HTTP as JSON. It is a Spring Boot application (Java 21) using `JdbcTemplate` against the same `coe_dept` database, including the `finance.get_outstanding_fees()` function. It serves two roles — **student** and **admin** — and the Project 1 front end runs entirely on it.

Browser → Next.js 14 front end (:3000) → **Spring Boot API (:8080)** → PostgreSQL
coe_dept



The API landing page at <http://localhost:8080/>.

2. Technology & Structure

Concern	Choice
Language / runtime	Java 21
Framework	Spring Boot 3.5 (Spring Web)
Database access	Spring JDBC (<code>JdbcTemplate</code>) — runs the Project 1 SQL directly
Password hashing	BCrypt (<code>spring-security-crypto</code>)
Authentication	Stateless JWT (HS256) with a role claim, verified by a servlet filter
Build	Maven

Layered controller → service → repository with dto, security and config packages.

3. API Endpoints

Student actions live under `/api/me/*`; department-admin actions under `/api/admin/*`. "write" endpoints change the database.

Method & path	Role	Purpose
POST <code>/api/auth/register</code> login	public	Account + JWT (login is role-aware)
GET <code>/api/me</code>	student	Profile + fee summary
PUT <code>/api/me/profile</code>	student · write	1 — edit contact details
POST <code>/api/me/payments</code>	student · write	2 — make a fee payment
GET <code>/api/me/registrations</code> available-courses	student	3 — registration data
POST / DELETE <code>/api/me/enrollments</code>	student · write	3 — register / drop a course
GET <code>/api/me/grades</code>	student	Completed results
GET <code>/api/admin/students</code>	admin	Class roster + outstanding (the JSON function)
GET <code>/api/admin/lecturers</code> course-options tas courses	admin	Option lists / catalogue
POST / DELETE <code>/api/admin/lecturer-course</code>	admin · write	4 — assign / remove lecturer→course
POST / DELETE <code>/api/admin/lecturer-ta</code>	admin · write	5 — assign / remove lecturer→TA
POST <code>/api/admin/courses</code>	admin · write	Add a course
GET <code>/api/fees/outstanding</code>	authenticated	Project 1 JSON function (all students)

4. Authentication & Authorization

- **Passwords** are BCrypt-hashed. **Login/registration** return a signed JWT (HS256) carrying the subject, email and **role**.
- A servlet filter protects every `/api/**` route except `/api/auth/**`; a missing/invalid token yields `401`.
- **Role-based access:** `/api/admin/**` requires an `ADMIN` token and `/api/me/**` a `STUDENT` token — a cross-role call returns `403`. Students may only read their own records (ownership check).
- **Validation:** payments must be positive and within the balance (`400`); a duplicate assignment or registration returns `409`.

5. Implementing the Project 1 Functions

The admin roster and `GET /api/fees/outstanding` both run the Project 1 function; writes use the same thin-repository style:

```
// FeesRepository.java (functionality 2)
public PaymentDto insertPayment(long studentId, BigDecimal amount, String method) {
    return jdbc.queryForObject(
        "INSERT INTO finance.payments (student_id, amount, method, reference) " +
        "VALUES (?, ?, ?, 'PAY-' || ? || '-' || to_char(now(),'YYYYMMDDHH24MISS')) " +
        "RETURNING payment_id, amount, paid_on, method, reference",
        PAYMENT_MAPPER, studentId, amount, method, studentId);
}

// AdminRepository.java (functionality 4)
public int addLecturerCourse(int lecturerId, int courseId, String year, int sem) {
    return jdbc.queryForObject(
        "INSERT INTO academic.lecturer_course_assignment " +
        "(lecturer_id, course_id, academic_year, semester) VALUES (?, ?, ?, ?) RETURNING"
        "assignment_id",
        Integer.class, lecturerId, courseId, year, sem);
}
```

5.1 The front end runs on this API

Class & Fees

70 students · balances from the `get_outstanding_fees()` database function.

Student ID	Name	Email	Billed	Paid	Outstanding
22384451	Abu Neaquittae Golda	22384451@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22376708	Adjei Priscilla	22376708@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22412982	Adu Mini	22412982@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22357814	Adzasa Stephen Yaw	22357814@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22375367	Afia Beaa Osei-Safo	22375367@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22397756	Agbemavi Ryan	22397756@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22369321	Agormeda Nathaniel Tetteh	22369321@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22301848	Ahmad Mohammed Sahih Kayelgu	22301848@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22339520	Amprofi Yaa Obeng	22339520@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22333597	Asante Esme Lilian	22333597@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22268986	Asante Gabriel Kwaku	22268986@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22381577	Botchway Daniel	22381577@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22315830	Brian Assibey-Yeboah	22315830@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22388189	Caleb Mensah	22388189@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22393520	Cyril Desmond Ofori	22393520@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22407018	Daniel Dwomoh Frimpong	22407018@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22312110	David Kwame Odoi-Anim	22312110@st.ug.edu.gh	GH¢12,300.00	GH¢9,000.00	GH¢3,300.00
22271867	De-Andra Rebecca Ayebo	22271867@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22300896	Doe Collins Kweku	22300896@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22402666	Dogbatse Darlington	22402666@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22397491	Douglas Kwaw Adjei	22397491@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22387715	Dzidzor Apu Apawudza	22387715@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382302	Edward Kakra Ankrah	22382302@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22382601	Eklou Juliet	22382601@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22379061	Emmanuel Akotuah Osae	22379061@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22368809	Emmanuel Dery	22368809@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22370498	Ethan Edric Kweku Nartey	22370498@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22328334	Etsey Hannah Seyram	22328334@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22321110	Gideon Nana Osei Amofa	22321110@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382425	Gilbert Akwasi Sarkodie Yeboah	22382425@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22396551	Jerrold Xornam Kyekye	22396551@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22398562	Joseph Amankwah	22398562@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22398596	Joshua Appiah	22398596@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22394866	Joshua Kwame Asirifi	22394866@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22385323	Jude Gyampoh Addo	22385323@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22415339	Kelvin Kwesi Saah	22415339@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22303421	Kemausuor Winambe Tetteh-Kumah	22303421@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22407033	Kenzi Segbefia	22407033@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22299189	Kessey Ntiako David	22299189@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22407837	Kingsley Caldicock Quartey	22407837@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22412615	Kofi Boateng Oware-Tano	22412615@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22411009	Kwaku Aninkorah Barimah	22411009@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22382547	Kwame Ayeh Obeng	22382547@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22373317	Kwamena Kesse Quaicoe	22373317@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22395074	Lydia Tiwaah	22395074@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22339058	Maame Abena Amihere Ahu	22339058@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22302628	Maame Araba Grant-Aidoo	22302628@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00

22396566	Manford Kelvin Oppong	22396566@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
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22325819	Nana Adwoa Dansowaah Odoom	22325819@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
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22306910	Newlove Yeboaah Kwarfo	22306910@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22385472	Obeng Ernest Antwi	22385472@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22399214	Obeng Ruth	22399214@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
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22263126	Owusu Koranteng Yaw Poku	22263126@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22373463	Owusu Nana Boadiwaa	22373463@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22306021	Paul Badu Amponsah	22306021@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00
22381702	Paula Akosua Asiedua Frimpong	22381702@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22387846	Quaicoo Emile	22387846@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22377537	Reuben Adomako	22377537@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22263922	Romel Alvin Nii Lartey Lartey	22263922@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
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22403781	Sekyere Kofi Bempong	22403781@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22304260	Tetteh Christian Edward Nii Mantey	22304260@st.ug.edu.gh	GH¢12,300.00	GH¢0.00	GH¢12,300.00
22304013	Tietaah Sonnu	22304013@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22416112	Troy Thomas	22416112@st.ug.edu.gh	GH¢12,300.00	GH¢12,300.00	GH¢0.00
22302188	Van Jerry Quansah	22302188@st.ug.edu.gh	GH¢12,300.00	GH¢8,000.00	GH¢4,300.00
22299949	William Enchill	22299949@st.ug.edu.gh	GH¢12,300.00	GH¢5,000.00	GH¢7,300.00

Admin Class & Fees — served by `GET /api/admin/students` , backed by the outstanding-fees function.

Lecturer & TA Assignments

Lecturer → Course (functionality 4)

Lecturer

Course

Semester

Select lecturer

Select course

2

Assign

Course	Title	Lecturer	Sem	
CBAS 208	Academic Writing I	Prof. Elsie Adjei	1	Remove
CPEN 202	Object Oriented Programming	Dr. Kwame Mensah	1	Remove
CPEN 204	Data Structures and Algorithms	Dr. Ama Boateng	1	Remove
CPEN 206	Linear Circuits	Dr. Yaw Owusu	1	Remove
CPEN 210	Digital Logic Design	Dr. Efua Danso	1	Remove
MATH 221	Calculus II	Mr. Kofi Antwi	1	Remove
CBAS 210	Academic Writing II	Dr. Ama Boateng	2	Remove
CPEN 208	Introduction to Software Engineering	Prof. Elsie Adjei	2	Remove
CPEN 212	Digital Systems Design	Dr. Efua Danso	2	Remove
CPEN 214	Signals and Systems	Dr. Yaw Owusu	2	Remove
CPEN 216	Electronic Circuits	Dr. Kwame Mensah	2	Remove
MATH 224	Linear Algebra	Mr. Kofi Antwi	2	Remove

Lecturer → Teaching Assistant (functionality 5)

Lecturer

Teaching Assistant

Course

Select lecturer

Select TA

Select course

Assign

Lecturer	TA	Course	
Dr. Ama Boateng	Selorm Agbeko	CPEN 204	Remove
Dr. Efua Danso	Linda Ofori	CPEN 212	Remove
Prof. Elsie Adjei	Akosua Frimpong	CPEN 208	Remove
Prof. Elsie Adjei	Nana Kwabena Asare	CPEN 208	Remove

Admin Assignments — the Assign/Remove buttons call the `/api/admin/lecturer-course` and `/api/admin/lecturer-ta` write endpoints (functionalities 4 & 5).

6. Non-Functional Requirements

Category	How the service addresses it
Security	BCrypt hashing; stateless HS256 JWT with a role claim; role-based authorization + ownership checks; parameterized SQL only; DB secret via <code>DB_PASSWORD</code> env var.
Validation & errors	A central <code>ApiException</code> + handler produces consistent JSON errors with correct codes (400/401/403/404/409).
Performance	HikariCP pooling; indexed lookups; a stateless design that scales horizontally.
Reliability & integrity	FK + UNIQUE constraints enforce integrity; <code>@Transactional</code> registration; deterministic, re-runnable SQL.
Maintainability	Clear controller/service/repository layering, DTO records, one responsibility per class.
Portability	Standard PostgreSQL and Maven; no hard-coded secrets; the JWT is a plain HS256 token any tool can read.

7. How to Run

1. Ensure `coe_dept` exists (Project 1 scripts or the backup).
2. `cd "PROJECT 2/api"`, set `DB_PASSWORD`, `mvn spring-boot:run` (port 8080).
3. Verify: log in (`POST /api/auth/login`) as a student or as `admin@coe.ug.edu.gh` , then call the matching `/api/me/*` or `/api/admin/*` endpoints, or `GET /api/fees/outstanding` .

Deliverables: Spring Boot source in `PROJECT 2/api` ; database scripts + backup in `PROJECT 2/database` ; full source on GitHub at <https://github.com/doxaodoi/project-1-2> .

Appendix — Running the Whole Stack with Docker

The entire system — database, API and frontend — is containerised so it can be started with a single command, with no local PostgreSQL, Java or Node installation required. From the submission root:

```
docker compose up --build
```

```
db (PostgreSQL) → api (Spring Boot :8080) → web (Next.js :3000)
```

What each service does

Service	Image / build	Role
db	postgres:16-alpine	On first start, automatically runs the Project 1 scripts <code>01_schemas</code> → <code>04_seed</code> to build and populate <code>coe_dept</code> .
api	built from <code>PROJECT 2/api</code>	Spring Boot service; reaches the database by the compose service name <code>db</code> .
web	built from <code>PROJECT 1/web</code>	Next.js 14 frontend; calls the API by the service name <code>api</code> . Open <code>http://localhost:3000</code> .

Composition (abridged `docker-compose.yml`)

```
services:
  db:
    image: postgres:16-alpine
    environment: { POSTGRES_DB: coe_dept, POSTGRES_PASSWORD: coe_dept_pw }
    volumes:
      - "./PROJECT 1/database/01_schemas.sql:/docker-entrypoint-initdb.d/01_schemas.sql:ro"
      - "./PROJECT 1/database/02_tables.sql:/docker-entrypoint-initdb.d/02_tables.sql:ro"
      - "./PROJECT 1/database/03_functions.sql:/docker-entrypoint-initdb.d/03_functions.sql:ro"
      - "./PROJECT 1/database/04_seed.sql:/docker-entrypoint-initdb.d/04_seed.sql:ro"
    healthcheck: { test: ["CMD-SHELL", "pg_isready -U postgres -d coe_dept"] }
  api:
    build: "./PROJECT 2/api"
    environment: { DB_URL: jdbc:postgresql://db:5432/coe_dept, DB_PASSWORD: coe_dept_pw }
    depends_on: { db: { condition: service_healthy } }
  web:
    build: "./PROJECT 1/web"
```

```
environment: { API_BASE_URL: http://api:8080 }  
depends_on: [api]
```

Why this matters (non-functional requirements)

- **Portability** — the system runs identically on any machine with Docker; no "works on my machine" setup drift.
- **Reproducibility** — the database is rebuilt deterministically from the same scripts every time, and `docker compose down -v` gives a clean slate.
- **Isolation & onboarding** — each tier runs in its own container on a private network; a new user is up and running with one command.

Sign in with the demo accounts documented earlier — student `22312110@st.ug.edu.gh` / `Student@123`, or admin `admin@coe.ug.edu.gh` / `Admin@123`.